

## Assignment III

In this assignment you will use regression models and compare them to neural network models. To do this, we will generate datapoints that follow the square function with some random noise added to each datapoint. The goal of the assignment is that you should understand the difference between different regression models and how they compare to a neural network model.

### Description

Implementation must be done in Python. You can use sklearn, keras and/or tensorflow to complete the assignment. The report should include information regarding the regression models, as well as the results that you get from step e. and f. You should also submit the code that you used as a .py or .ipynb file.

### Steps

- a. Generate data points in Python using:

```
x_data = np.linspace(-0.5, 0.5, 200)[: , np.newaxis]
noise = np.random.normal(0, 0.02, x_data.shape)
y_data = np.square(x_data) + noise
```

- b. Establish a linear regression model to predict y data (you can use sklearn).
- c. Establish a polynomial model with highest power of 2 to predict y data (you can use sklearn).
- d. Establish a three-layer (including input, 1 hidden (6 nodes) and output layer) neural network to predict y data (keras or tensorflow is recommended). Split the dataset into training 80% and testing data 20% using the following code in Python:

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x_data, y_data,
test_size = 0.2, random_state = 0)
```

- e. Calculate and compare mean squared errors of three models in (b), (c) and (d).
- f. Plot data points and curve of predictions in (b) and (c).

### Evaluation

The assignment is graded with U or G. It is important to note that this is an individual assignment.

This means that everything that you submit for grading must be created by you.

Plagiarism is not allowed in any form.